Q4

a. Let's assume a normal queue that can accommodate maximum six elements. At a given instance, front=2 and rear=4 and the queue elements look like this:

—, —, L, M, N, —

What will be the value of the front and rear if 'O' is inserted into the queue. Explain your answer

The queue will become —, —, L, M, N, O

~~Because the 'O' is inserted at~~

Because insertion occurs at a position following the rear

front x x x x x rear

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b. For the following code snippet what will be the output? Explain your answer.

```
#include<iostream>
#include<queue>
using namespace std;
int main(){
    queue<int> q;
    for(int i=1; i<=5;++i){
        q.push(i);              // now queue is [1,2,3,4,5]
    }
    cout<<"Initial size:"<<q.size()<<endl;
    for(int i=0;i<3;i++){   first time i=0
        q.push(q.front());  // [1,2,3,4,5]→push1→1,2,3,4,5,1→pop 2,3,4,5,
        q.pop();            // i=1⇒ [3, 4,5,1,2]
    } cout<<q.front()<<" ";  // i=2⇒[4,5,1,2,3]
    cout<<"\nSize at end: "<<q.size()<<endl;
    return 0;
}
```

Output:

Initial size: 5

4

Size at end: 5